

Experiment 3: DC-DC Converter for Maximum Power Point Tracking (MPPT) and Charge Control

Part 1: Open-Loop Buck DC-DC Converter

ECEN 5517-012

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BENCH 5

I. Introduction

The goal of this experiment is to design, construct, and test a DC-DC buck converter to charge a battery. We will also implement a duty cycle control for the converter to allow for maximum power point tracking (MPPT) in the second part of the experiment.

II. Experimental Procedure and Results

a. Inductor Design and Construction

We designed for a target ripple of 10% at a switching frequency of 100 kHz. At a peak power of 85W from the source, a battery/inductor current average of 6.54A is expected. A 10% ripple of that is 654 mA. Peak power delivery will occur at $V_{mpp} = 17.2V$ and 13 V battery voltage. The duty cycle, nominally, would be 0.756 for these voltage values in a lossless case. Thus, using the following equation taken from analyzing buck converter waveforms, we will be designing for an inductance of **24.287 uH**:

$$L = \frac{(V_g - V_{batt})DT_s}{2\Delta i_L}$$

Next, via the Kg method to organize our constraints, we arrive at a Kg where **PQ26/25** is the smallest core available that satisfies our needs. Next the number of turns was calculated using the maximum B field of 0.35T for the N95 ferrite core per the TDK Datasheet (larger than N87 due to larger permeability), a maximum current of 7.19 A, and the bobbin parameters.

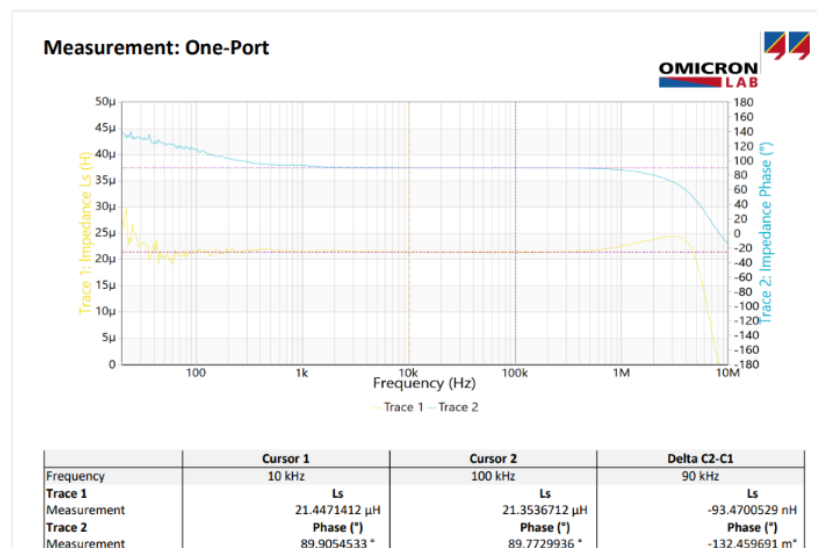
$$n \geq \frac{L \cdot I_{max}}{B_{max} \cdot A_c} = 4.23 \rightarrow \mathbf{5 \text{ turns}}$$

Using this number of turns, we can calculate the last two parameters: the air gap and wire gauge size. The equations for both are shown below. A K_u of 0.5 is assumed based on the inductor's size.

$$l_g = \frac{\mu_o A_c n^2}{L} = \mathbf{0.006 \text{ in.}}$$

$A_w \leq \frac{K_u W_a}{n} \approx 5 \text{ mm}^2$, 11 AWG is the largest we can go, the largest in the lab was **12 AWG**, so we will use that.

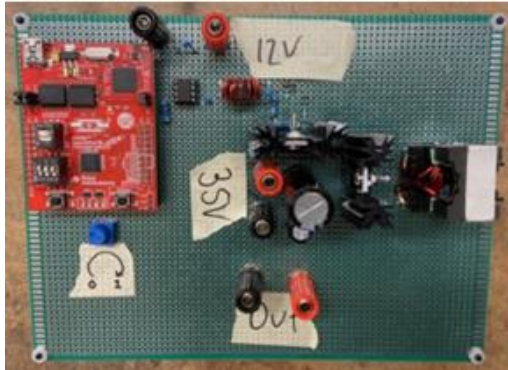
The airgap was created with one **3-mil shim on both sides of the core**. The measured inductance with the Bode Analyzer was **only 21.35 uH**, as shown in the report below at 100 kHz:



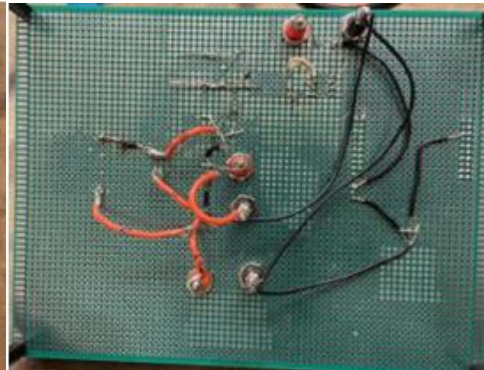
It also had a series resistance of 1.66mΩ with a 4-wire test as well before being soldered.

b. Buck Power Stage Construction

Below are pictures of our constructed buck converter. The MOSFET, diode, and power stage input capacitor are clustered together to minimize the size of their loop, and the two power stage capacitors are aligned to create a star ground. Since the MOSFET is at the top of the arrangement, we can build the gate driver circuit immediately above it to minimize the distance between it and the gate.



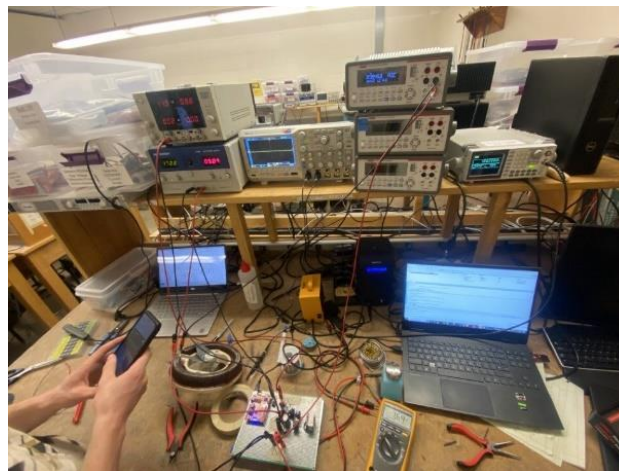
Top of Buck Converter



Bottom of Buck Converter

c. Buck Power State Testing

After quickly confirming that the gate drive circuit was operating correctly, we set up the bench to be able to test the power stage of the buck converter. Our two DC power supplies are set to 12V and 17.2V for the gate driver and power input, respectively.



Step 3 Test Setup

The power input and 12V inputs are driven by two independent power supplies that track current draw as well as voltage supplied. We used these values for our values of input power. The output is connected in series to an ammeter and rheostat and in parallel to a voltmeter. We got roughly 85 W of power output at a duty cycle of 79.49% and took the following waveforms and data at those parameters. We adjust the rheostat and potentiometer controlling duty cycle in tandem to reach 13V out AND 85W out at the same time.

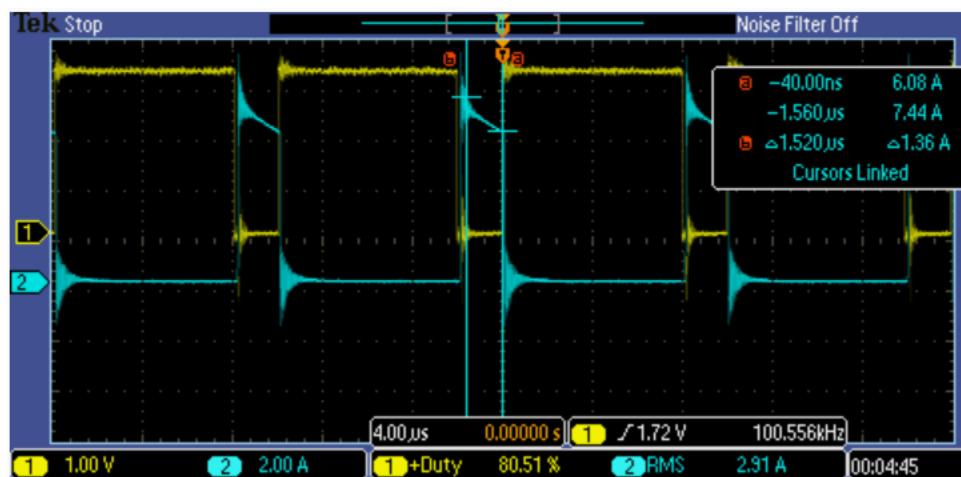
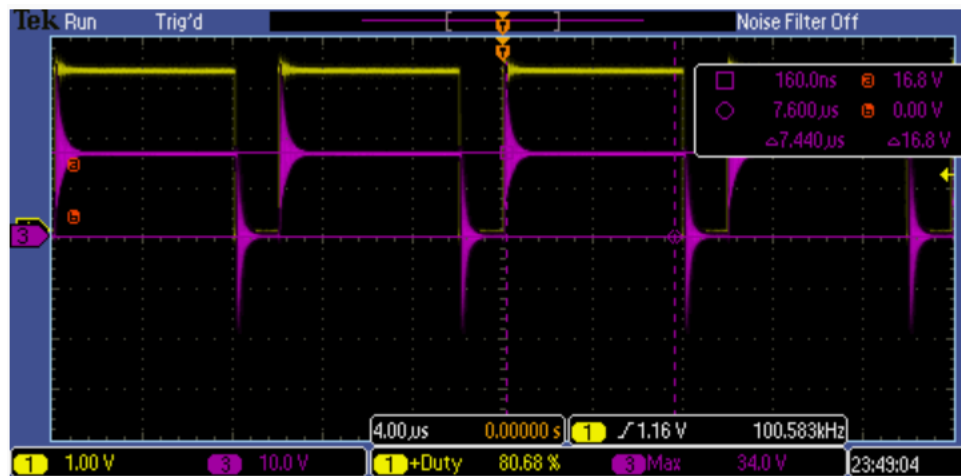
i. *Input and Output DC Voltages and Currents, Efficiency:*

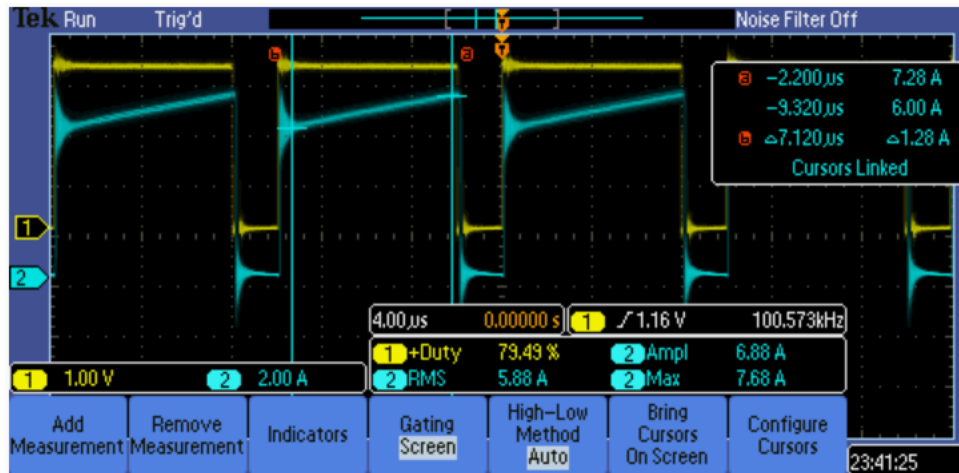
	Voltage (V)	Current (A)	Power (W)
Source Power Supply	17.19	5.35	91.97
Load	12.99	6.65	86.38

This result gives a converter **efficiency of 93.93%**, which is quite respectable for a single-transistor converter. It also required an initial **duty cycle of 79.49%**.

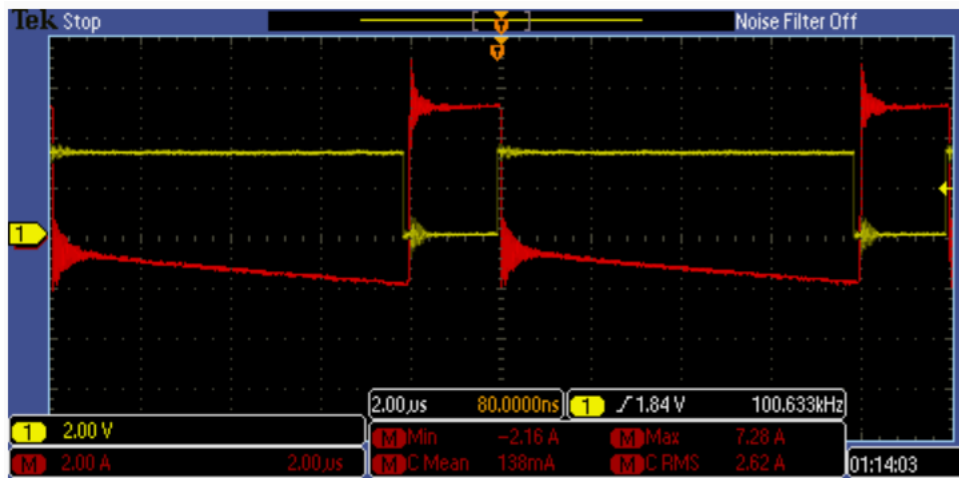
ii. *Labeled waveforms for $V_D(t)$, $I_T(t)$, $I_D(t)$, $I_L(t)$, $I_{C1}(t)$, $V_{C1}(t)$*

Each of the graphs below show the gate driver duty cycle in yellow.

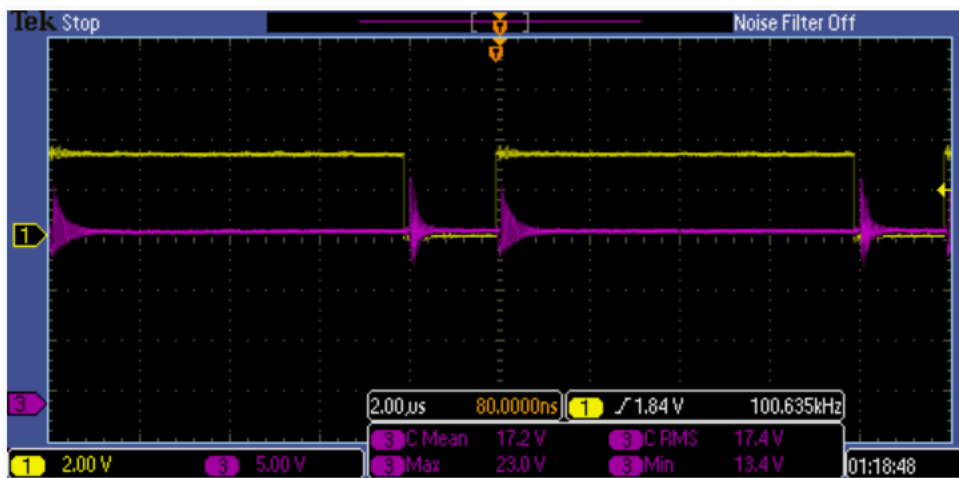




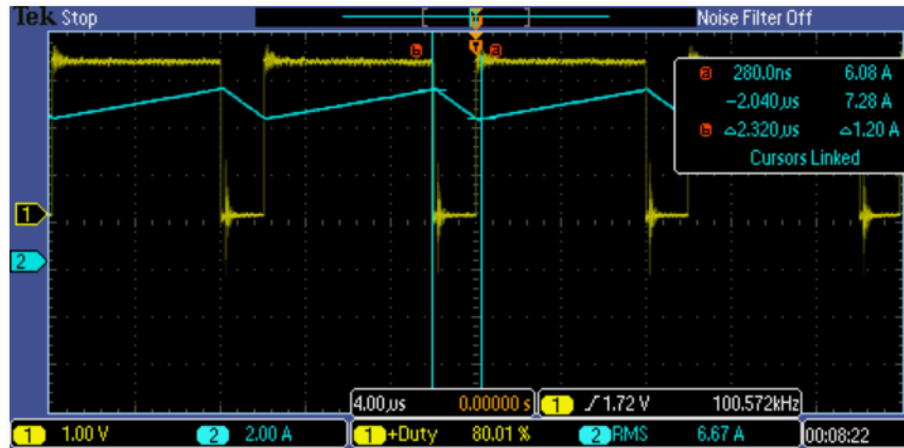
Time: 4 μ s/div Cyan: $I_T(t)$ 2A/div, Yellow: J6.1 Duty Cycle 1V/div



Time: 2 μ s/div Red: $I_{C1}(t)$ ($I_g(t) - I_T(t)$) 2A/div, Yellow: J6.1 Duty Cycle 2V/div



Time: 2 μ s/div, Purple: $V_{C1}(t)$ 5V/div, Yellow: J6.1 Duty Cycle 2V/div



Time: 4us/div Cyan: $I_L(t)$ 2A/div, Yellow: J6.1 Duty Cycle 1V/div

iii. Comparing Measured Inductor Ripple vs Designed-For Inductor Ripple

From the cursors shown in the inductor current figure above, our ripple is 0.6 A. We can use the upper and lower cursor to find the average inductor current of 6.68A. **This corresponds to an inductor current ripple of 8.98%**, quite close to the 10% current ripple we designed for in the inductor design stage.

iv. Estimating ESR of Capacitor C1 and Comparing to Data Sheet Value

From the cursor measurements in the V_{C1} and I_{C1} waveforms in II.c.ii (the 4th and 5th waveform plots), we found that the cycle mean and RMS voltage of C1 was 17.2 and 17.4 V, respectively. Additionally, the I_{RMS} current of the capacitor is 2.62 A. The rough ESR calculation using these values is shown below, in which subtracting the mean from the rms gives a rough estimate of the AC rms voltage across the capacitor.

$$\frac{V_{rms} - V_{mean}}{I_{rms}} = 76.34\text{m}\Omega$$

This result is much higher than the datasheet impedance of 13 milliohms, which is perhaps a lossless $\sqrt{L_{res}/C_{res}}$ impedance metric at self-resonance, we couldn't figure out what frequency exactly. Forums on electrolytic ESR tend to put the value at roughly 50mΩ to 100mΩ for similarly sized and rated aluminum electrolytic capacitors at 20C. **The datasheet also gives an RMS ripple rating of 4.08 A, which our capacitor operates well below at 2.62A RMS.** The power dissipated by our capacitor using the ESR estimate and I_{RMS} is shown below.

$$P_C = I_{RMS}^2 ESR = 0.524 \text{ W}$$

0.524 W accounts for ~0.6% of the total 91.97W supplied by the 17.2V power supply. However, if we assumed the AC component to be purely sinusoidal with the capacitor being a constant DC source, the actual AC ripple of the signal would be 3.81V. The rms voltage divided by the rms current then of the DC blocked signal would be 1.0283Ω. This figure seems less aligned with both the datasheet and the fact that 6.9W would have been dissipated across the capacitor, higher than total loss we encountered. Thus, we explored this alternative approach because divorcing AC rms from AC+DC rms is usually not as simple as just subtracting the mean of the AC+DC signal, but here we have no other way of truly validating.

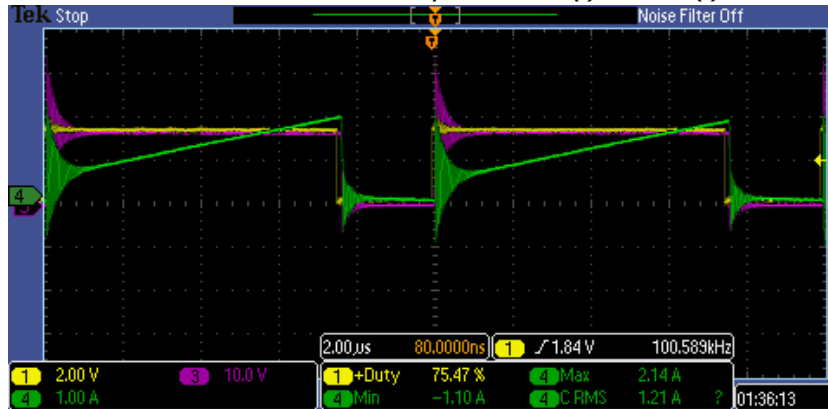
d. Load Test

i. Load Test Data at 15W Output

	Voltage (V)	Current (A)	Power (W)
Source Power Supply	17.23	0.890	15.3347
Load	13.03	1.15	14.9845

This results in an **efficiency of 97.64% with a duty cycle required of 76%**. This is only a 3% difference in duty cycle from the previous 85W test.

ii. Labeled Waveforms for 15W Output for $V_D(t)$ and $I_T(t)$



Time: 2us/div **Green:** $I_T(t)$ 1A/div, **Purple:** $V_D(t)$ 10V/div **Yellow:** J6.1 Duty Cycle 2V/div

iii. 15W Output Discussion

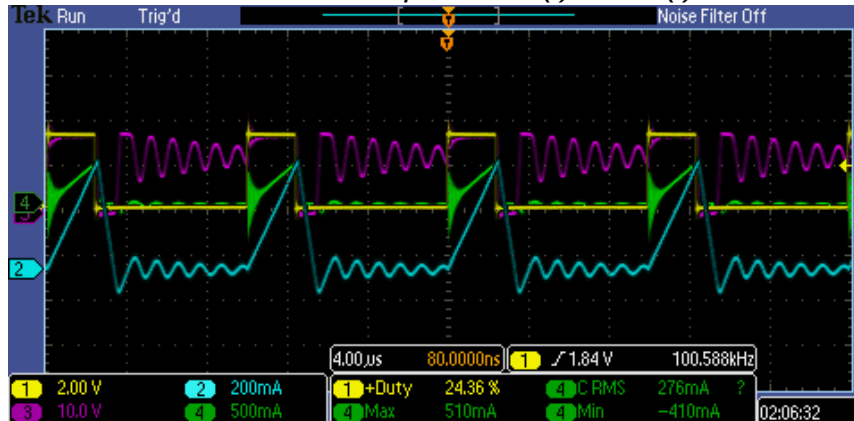
Clearly, transistor current RMS had gone down considerably compared to 85W and inductive noise manifested in sudden changes in current through the diode is seen in the voltage spikes in $V_D(t)$ as well as capacitive ringing in the transistor current $I_T(t)$.

iv. Load Test Data at 1W Output

v.	Voltage (V)	Current (A)	Power (W)
Source Power Supply	17.23	0.064	1.10272
Load	12.96	0.08	1.0368

This results in an **efficiency of 94.02% with a duty cycle required of 24.2%**. We have now reached DCM territory in which the conversion ratio to reach 13V at the output changes from $V = DV_g$ to a DCM $M(D,K)$.

vi. Labeled Waveforms for 1W Output for $V_D(t)$ and $I_T(t)$



Time: 4us/div **Green:** $I_T(t)$ 0.5A/div, **Purple:** $V_D(t)$ 10V/div **Yellow:** J6.1 Duty Cycle 2V/div
Cyan: Inductor Current: $I_L(t)$ 0.2A/div

vii. 1W Output Discussion

As mentioned, the converter is operating in DCM. Inductor current was included to show that the diode junction capacitance and inductor will pick up the harmonic of the piecewise inductor current signal that satisfied $f = \frac{1}{2\pi\sqrt{LC_{diode}}}$ and resonate. Indeed, diode voltage and inductor current are in quadrature and decay with the same lossy exponential attenuation as the load (battery) connected in series.

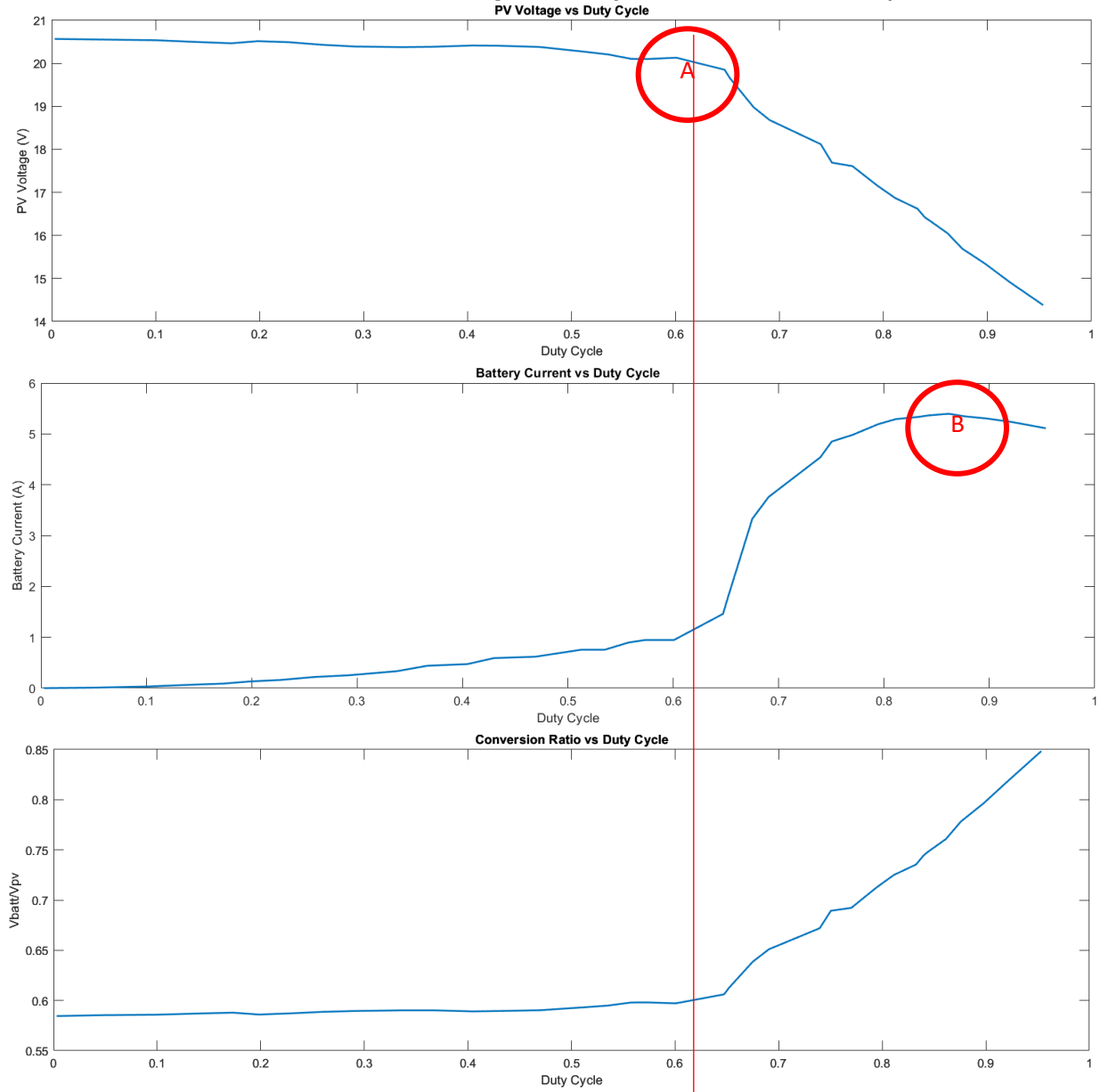
e. Open-Loop Behavior in PV System

Measured PV Panel Data vs Duty Cycle

Time: 1:06PM 2/11/24 to 2:19PM 2/11/24 Irradiance: 1130 W/m ² to 1107 W/m ² *Peak Power Data Points*								
Duty Cycle (%)	PV Voltage	PV Current	PV Power	Battery Voltage	Battery Current	Battery Power	Conversion Ratio (V _{batt} /V _{pv})	Efficiency (P _{batt} /P _{pv}) *Peak Eff*
0.3	20.566	0.003	0.061698	12.02	0	0	0.584459788	0
4.823	20.552	0.008	0.164416	12.03	0.009	0.10827	0.585344492	0.658512554
10.03	20.536	0.022	0.451792	12.03	0.0312	0.375336	0.585800545	0.830771683
14.05	20.494	0.042	0.860748	12.03	0.0649	0.780747	0.587001073	0.907056421
17.3	20.465	0.056	1.14604	12.03	0.089	1.07067	0.587832885	0.934234407
19.84	20.514	0.086	1.764204	12.02	0.1315	1.58063	0.585941308	0.89594514
22.83	20.49	0.103	2.11047	12.03	0.162	1.94886	0.587115666	0.92342464
26.01	20.43	0.137	2.79891	12.026	0.2202	2.6481252	0.588644151	0.946127314
29.15	20.39	0.157	3.20123	12.02	0.252	3.02904	0.589504659	0.9462113
33.78	20.376	0.209	4.258584	12.025	0.333	4.004325	0.590155084	0.940294943
36.66	20.384	0.265	5.40176	12.03	0.44	5.2932	0.59016876	0.979902846
40.47	20.413	0.294	6.001422	12.025	0.4726	5.683015	0.589085387	0.946944741
42.98	20.408	0.37	7.55096	12.03	0.5907	7.106121	0.589474716	0.941088418
46.92	20.38	0.39	7.9482	12.029	0.6177	7.4303133	0.590235525	0.934842266
51.26	20.268	0.482	9.769176	12.023	0.755	9.077365	0.593201105	0.929184304
53.51	20.206	0.487	9.840322	12.02	0.756	9.08712	0.59487281	0.923457586
55.76	20.102	0.579	11.639058	12.02	0.897	10.78194	0.597950453	0.926358473
57.34	20.099	0.607	12.200093	12.02	0.947	11.38294	0.598039703	0.933020756
60.07	20.13	0.607	12.21891	12.02	0.947	11.38294	0.597118728	0.931583914
64.72	19.85	0.97	19.2545	12.03	1.46	17.5638	0.60604534	0.912191955
65.21	19.66	1.19	23.3954	12.045	1.8	21.681	0.61266531	0.926720637
67.5	18.98	2.3	43.654	12.121	3.33	40.36293	0.6386196	0.924610116
69.05	18.68	2.83	52.8644	12.16	3.76	45.7216	0.650963597	0.864884497
73.96	18.12	3.354	60.77448	12.176	4.54	55.27904	0.67196468	0.90957652
75.04	17.69	3.746	66.26674	12.195	4.85	59.14575	0.689372527	0.892540511
77.01	17.61	3.88	68.3268	12.19	4.98	60.7062	0.692220329	0.888468361
79.44	17.15	4.206	72.1329	12.22	5.19	63.4218	0.712536443	0.879235411
81.11	16.87	4.345	73.30015	12.23	5.29	64.6967	0.724955542	0.882627116
83.26	16.62	4.458	74.09196	12.223	5.333	65.185259	0.73543923	0.879788563
83.97	16.42	4.53	74.3826	12.221	5.355	65.443455	0.744275274	0.879822095
84.34	16.36	4.56	74.6016	12.23	5.365	65.61395	0.747555012	0.879524702
86.15	16.05	4.682	75.1461	12.212	5.395	65.88374	0.760872274	0.876741973
87.58	15.69	4.762	74.71578	12.21	5.346	65.27466	0.778202677	0.873639544
89.78	15.34	4.826	74.03084	12.217	5.301	64.762317	0.796414602	0.874801866
92.16	14.91	4.88	72.7608	12.21	5.232	63.88272	0.818913481	0.87798265
95.35	14.38	4.936	70.97968	12.2	5.11	62.342	0.848400556	0.878307707

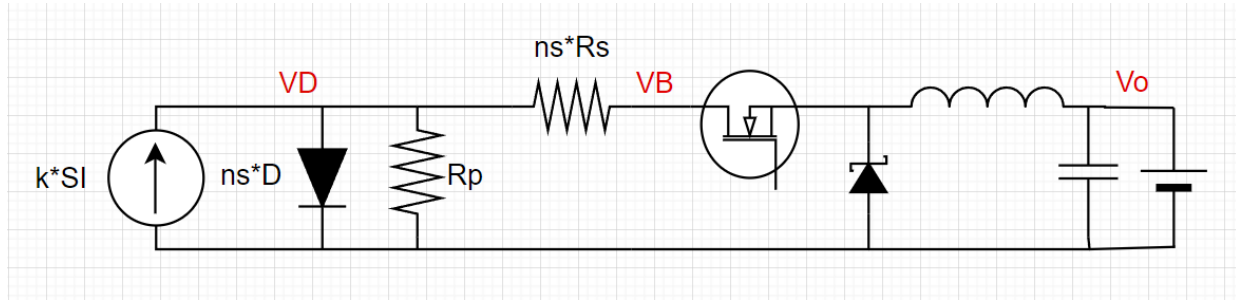
f. Data Analysis and Simulation

i. Labeled Plots of PV Voltage, Battery Current and V_{batt}/V_{pv}



ii. Theoretical Explanation of Salient Features from Real World Data

First, feature **A** will be explained. When the output voltage of a buck converter is fixed and the input voltage is not fixed, duty cycle actually will change the input voltage. This is the case with the PV panel model in which we have a current source in parallel with a diode. For a buck, the conversion ratio is $V_o = D V_g$. So if V_o is fixed at 12V and the duty cycle is 10%, V_g would theoretically have to be 120V (an ideal buck converter backwards is a boost converter when just looking at voltage and not power flow). At such a voltage, the 36 PN junction cells in series would forward bias not actually send power to the buck converter, instead it is all dissipated as heat in the panel itself. The typical characteristics will occur just as soon as the panel voltage is below roughly $0.6V_{fw} * 36 = 21.6V$. A script in appendix A was created to estimate when this should happen using the lossy model and it will be used to compare to the actual model:

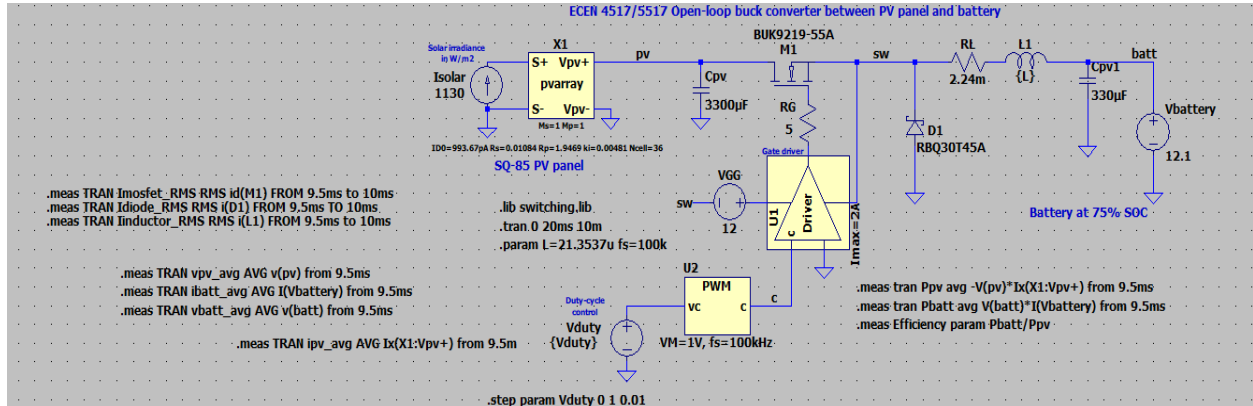


Assuming $R_s = 10.84\text{m}\Omega$, $R_p = 1.9469\Omega$, $k \cdot S_I = .00481 \cdot 1114\text{W/m}^2 = 5.36\text{A}$
 And also incorporating known losses: $R_{dsON} = 0.034\Omega$, $V_{diode} = 0.47\text{V}$, $R_{ind} = 0.00166\Omega$:
 With $V_{batt} = 12.1\text{V}$ on average. And $V_{Don} = V_{oc} = 21.6\text{V}$, assuming the current at the duty cycle where the diodes are no longer forward biased is still 0, gives a predicted “normal” operating region for all duty cycles **above 57%**. In real life, the forward voltage model of the diode is too ideal and the precise start of the linear region is more nebulous, roughly a little after 0.6-0.65. Around 0.58 according to our LTSpice model. See Appendix A for script used to predict this.

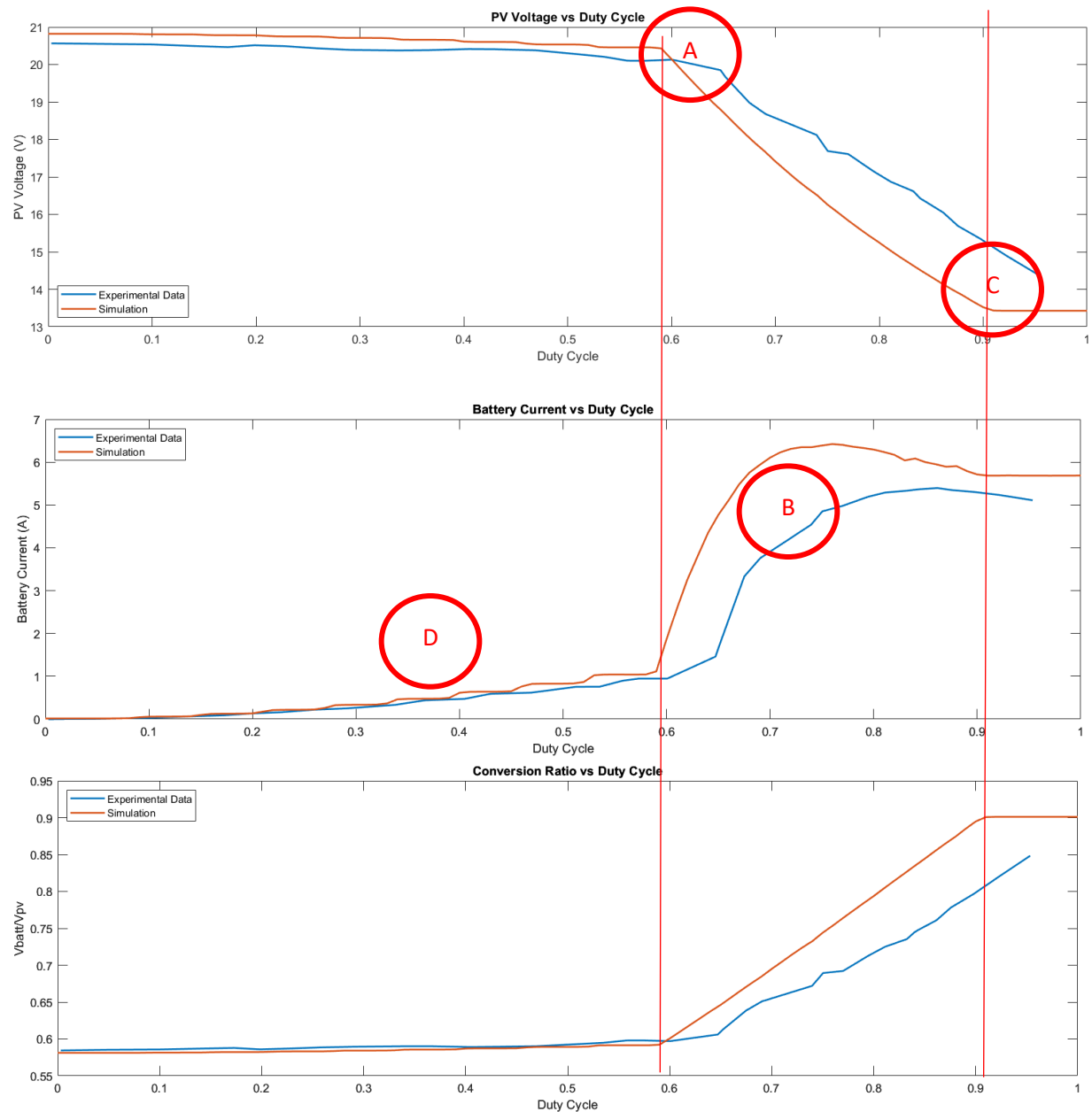
For a fixed output voltage too, we know that the battery current will be directly proportional to the power the panel is supplying. Once the panel is operating at MPP, the battery current will be maximized, which occurs in our real-life test at around 86% which is what feature **B** is. This is way higher than the expected value from LTSpice below which occurs at around 76% duty cycle, which is what previous models also expected and what we would expect from our benchtop tests in II.c.i.

iii. LTSpice PV Voltage, Battery Current and V_{batt}/V_{pv} Plots

We reproduced our experiment in simulation with LTSpice. We updated the panel and converter model from Experiment 1 with panel parameters extracted in Experiment 2, and used a parameter sweep of the duty cycle to mimic our experiment. The battery is modeled as a voltage source with the average value of the experimental data. The LTSpice circuit model is shown below. The $2.24\text{m}\Omega$ is the 4-wire measurement of the inductor from its soldered leads so we used that as a worst-case scenario in this model instead of the $1.66\text{m}\Omega$ in II.a.



Data was collected from this simulation by averaging each measurement (I_{batt} , I_{PV} , V_{PV} , and V_{batt}) from 9.5 ms to 10 ms at each 1% step in duty cycle then processed using MATLAB. The results are shown below.



Salient features **A** and **B** have already been discussed in the II.f.ii discussion. There are two more salient features worth mentioning that come to light in our simulation. The first is a second cutoff (**C**) that occurs at a fairly high duty cycle, close to 90%, and the system is effectively converted to a DET system. This is caused simply by the PWM module within the LTSpice schematic which is parametrically constrained to interpret all V_{duty} signals greater than or equal to 0.9 to be 0.9. Otherwise, the LTSpice schematic would've continued along its linear trajectory. The final salient feature is **D** which shows a step-like incline on battery current as a function of duty cycle when the panel's PN junctions are forward biased. This is due to the exponential nature of the IV curve in this forward biased condition. Using a simple "on" forward voltage model would make it so that the curve before **A** would be totally flat but no diode truly behaves that way. The discrete stepping behavior is a little more inexplicable though.

iv. Comparison of Measured vs Simulated Plots

The measured and simulated plots compare very well. Both of them have an inflection point right around a duty cycle of 60%, with this turn happening at about a 5% lower duty cycle in the simulation. The PV voltage and conversion ratio plots are both significantly smoother in the simulation, and the battery current is also much higher with panel voltage supposedly falling much steeper than we observed past the forward bias cut off (A). Overall, each plot has a very similar shape to its real-world counterpart, so we can conclude that practical loss elements are responsible for the discrepancy between the model and the collected data.

v. Optimum Duty Cycle and Maximum Power Based on Measured Data

We found that the maximum power point occurred at 86.15% duty cycle, where 75.15 W from the panel conferred 65.88 W to the battery, yielding an efficiency of 87.67%.

vi. Comparison between MPPT and DET

The DET from Experiment 2 was at 12.91 V and 5.244 A, meaning the battery charging power was 67.7 W, slightly higher than the 65.88 W with our MPPT circuit. This is unexpected, but not altogether inexplicable. Firstly, the irradiance had dropped to 1107 W/m² by the time we were taking our last data points, which is slightly lower than the 1114 W/m² we enjoyed during Experiment 2. The panel also could have been positioned less optimally, since our MPPT data was taken later in the afternoon when the sun's altitude was changing faster. Secondly, the battery could have been at a different state of charge in each experiment, meaning its IV charging behavior might be much different. The DET test from Experiment 2 was also measuring the voltage of an unloaded battery directly connected that was at around 12.2V when loaded right before. Our battery at the time of this lab had been loaded with a lightbulb, 3 multimeters, an oscilloscope, and two laptops charging, dropping the battery voltage considerably. By the time we started testing, our open circuit voltage was near 12.4V and we never got a measurement for what the unloaded charging voltage might be, but it would've probably been in the same range as the DET test. It is worth noting that the overall efficiency of our converter setup is still higher even with all the aforementioned skews in our data that would suggest the DET performed better. We only achieved a peak power of 75W from the panel when testing with the converter whereas in experiment two, we had reached almost 80W. This is an 87.67% efficiency with the converter vs an 84.6% efficiency in DET with skewed values.

III. Appendix

a. II.f.ii script

```
% assume Is, Id0, Rs and Rp have been extrapolated from
% Voc, Isc, Vmpp, Impp
clear all
clc
syms D VD Vdiode IL Rs Rp Is Vt L C Vbatt Ibatt Id0 ns Rdiode Rind Rfet1 Rfet2 Rsense
Vt = 26*10^-3
Ipv_generally = Is - Id0*(exp(VD/(ns*Vt))-1) - VD/Rp/ns;
% lets do it incorporating any losses that may arise too:
VB = VD - IL*Rs
VL1 = VB-Vbatt - IL*Rfet1 - IL*Rind - IL*Rsense
IC1 = IL - Ibatt
Ipanel1 = IL

VL2 = -Vbatt - Vdiode - IL*Rdiode - IL*Rind
IC2 = IL-Ibatt
Ipanel2 = 0

% now inductor VSB and cap CSB
eqn1 = D*VL1 + (1-D)*VL2 == 0
eqn2 = D*IC1 + (1-D)*IC2 == 0
Ipanel_avg = D*Ipanel1 + (1-D)*Ipanel2

IL_s = simplify(expand(solve(eqn2, IL)))
eqn1 = subs(eqn1, IL, IL_s)
Ipanel_avg = subs(Ipanel_avg, IL, IL_s)
VD_s = simplify(expand(solve(eqn1, VD)))
%lets make it dummer to understand: this is the
%value of VD given a Vbatt and D
VD_s_dummy = simplify(expand(subs(VD_s,[Rdiode Rind Rsense Vdiode Rfet1],[0 0 0 0])))
% we see VD_s will want to be driven into Vbatt/D.
% now in the case of the diodes in PV panel, at 0.6*36 = 21.6V,
% if VD_s is supposed to be > 21.6, then all the current will short
% through the panel diodes. And I panel will be 0. As it hits the boundary
% though, the diode only kind of turns on an there is some current that can
% go through. This current is DIL and so IL = Ileftover/D. That current
% times battery voltage is output power. We can plug and chug the dirt into
% the actual VD_s (more accurate line 27) soon for the report.
D_knee = solve(VD_s == 20.92, D)
D_knee = subs(D_knee, [Vbatt Rdiode Vdiode Rind Rs Rp Rfet1 Rsense], [12.1 0 0.47 0.00166 36*0.01084 1.9469 0.034 0])
```